Development of a Waterproof EMG Sensor for Improved Signal Quality by Using Load Balancing

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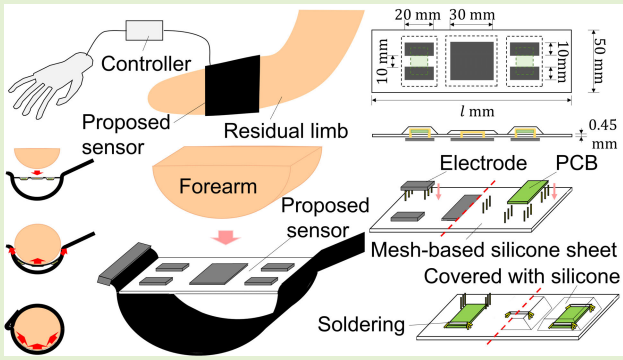
***Abstract*—Our purpose is the development of an electromyogram (EMG) sensor that can be used underwater. EMG signals are incorrectly measured when water flows between the electrodes and the skin, preventing the use of myoelectric prosthetic hands underwater. Conventional waterproof sensors mainly focus on the waterproofness of the electrode and do not consider the stability of contact between the electrode and the skin. Thus, if water enters the area between the electrode and the skin, the acquired signal becomes unstable. We propose an EMG sensor system that uses a flexible wearable band to prevent water from entering between the electrode and the skin and stably measure EMGs. The waterproofness of the sensor is ensured by attaching the EMG sensor to the mesh-based silicone sheet and covering the sensor with silicone. Moreover, waterproofness between the skin and the sensor is ensured by applying tension to the flexible sensor band while the band deforms along the forearm, and the mesh-based silicone sheet around the electrode closely attaches to the skin. The evaluation results regarding the signal quality of the signal to noise ratio (SNR), median frequency (MDF), and attenuation rate in air and water confirmed that the same EMG signals could be acquired in both environments. These results indicate that we have succeeded in developing a sensor that can measure muscle activity in water, as the pressurization caused by the band enhances the sensor's waterproofness.**



***Index Terms*—Conductive silicone, electromyogram (EMG) sensor, flexible sensor, myoelectric prosthetic hand, waterproof.**

I. Introduction

A. Background

MUSCLE contraction in humans is caused by electrical signals sent from the brain to the muscles via nerves. The signals generated to cause muscle contraction are called electromyogram (EMG) signals, from which the intended motion can be estimated. Controlling a myoelectric prosthetic hand, which is operated based on the wearer's intentions, is a typical application of this technology.

Various EMG sensors have been studied to improve the performance of myoelectric prosthetic hand control. EMG sensors are primarily divided into two types: invasive and noninvasive. Invasive sensors measure muscle contraction signals directly using needles or wires, which provide clean signals but also injure the body. Conversely, noninvasive sensors measure signals from the surface of the body and are further divided into two types: dry or wet, depending on the electrodes [1], [2]. Wet-type electrodes use a conductive paste between the skin and the electrode, which provides stable measurements; however, they cause discomfort and itching, and the running costs are high because they are disposable. Dry-type electrodes are made of metal, metal cloth, or conductive rubber, and are generally unstable because of the tendency of the electrode to slip on the skin. In addition, electrodes made of metal or metal cloths can cause pain or allergic reactions [3]. Electrodes made

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Please see the Acknowledgment section of this article for the author affiliations.

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TABLE I
COMPARISON OF CONVENTIONAL STUDIES AND PROPOSED MECHANISM

	Previous study of waterproof sensor				Basic research on flexible sensors used in this study			This study
	Iizuka[9]	COMETA[10]	Duan[11]	Ohtsu[12]	Jiang[7]	Mouri[4]	Togo[5], Jiang[8]	Proposed mechanism
Electrode material	Ag/AgCl	Ag/AgCl	Silicone	Silicone	Polypyrrole	Silicone	Silicone	Silicone
Equipment	Waterproof film presses down on electrode	Waterproof film presses down on electrode	Rubber band presses down on electrode	Rubber band presses down on electrode	Rubber band with electrodes sewn on to press down on electrode	Silicone sheet with electrode embedded to wrap	Rubber band presses down on electrode	Mesh-based silicone sheet with electrode embedded to wrap
Distributed of pressure	×	×	×	×	×	✓	×	✓
Waterproof electrode	✓	✓	✓	✓	×	×	×	✓
Waterproofing of entire sensor system	×	✓	×	×	×	×	×	✓
Waterproofing between skin and electrode	Waterproof film	Waterproof film	Electrode close contact with skin	Electrode close contact with skin	×	×	×	Mesh-based silicone sheet close contact with skin
Ability to follow changes in skin shape caused by muscle bulges	×	×	✓	✓	✓	✓	✓	✓

of conductive rubber must have stable electrical properties and flexibility to adhere to skin.

A wet-type electrode attaches to the skin via gel. In contrast, a dry-type electrode attaches to the skin by being pressed against it. However, if the pressure applied to the skin is uneven, the impedance between the skin and the electrode various, making it difficult to obtain normal EMG signals. Therefore, when measuring EMG signals using dry-type electrodes, ensuring even pressure of each electrode is important.

As described above, each type of EMG sensor has its disadvantages. To address this problem, in previous studies, our group developed dry-type electrodes with two-layered conductive silicone, a type of conductive rubber [4], [5], and confirmed comfortable wear and stable measurements.

Myoelectric prosthetic hands are required to improve a user's quality of life in activities of daily livings (ADLs). According to Linacre et al. [6], ADLs are classified into self-care, sphincter control, mobility, locomotion, communication, and social cognition, including many water-related tasks. In particular, many water-related activities make it difficult for others to intervene for privacy reasons; these include eating, bathing, and toileting. Therefore, myoelectric prosthetic hands need to be waterproofed. It is important for electric components to avoid water, but in terms of sensors, it is also necessary to consider the stability of the interface between the skin and an electrode in water.

In this study, we developed a waterproof EMG sensor. Waterproofing of EMG sensors is not just a matter of isolation from water; pressure loads must also be considered. The EMG sensor must be attached to the skin with an evenly distributed load because the characteristics at the skin-electrode interface are changed by the load; an uneven load allows water ingress at low pressure and affects the quality of the signal. Therefore, the design of a waterproof EMG sensor must include not only the electrode, but also the attachment method and peripheral mechanisms. The relationship between the peripheral mechanism and the signal quality in water is discussed in Section II.

B. Objective

The objective of this study was to develop a waterproof EMG sensor with a silicone sheet that enables stable measurement without water intrusion, even when the contact surface is deformed or the pressure changes, such as during muscle bulging. In particular, we applied an integrated sensor band using a mesh-based silicone sheet to disperse pressure, thereby solving the problem of poor signal quality due to water intrusion, which has hindered conventional sensors. The curvature of the skin changes greatly due to the movement of muscles, and the metal electrodes cannot respond to these changes. Hence, electrodes move away from the skin and water enters the gap between them. On the other hand, if the sensor has the form of a flexible band, such as one made of rubber, the pressure inside the band is constant, and a uniform force is applied to the skin; thus, these problems can be solved. Additionally, previous research has verified the performance of the circuits and flexible sensors being used here [4], [5], [7], [8], so this research focused on evaluating waterproofing only. Therefore, this study verified whether the quality of the signal changes under different conditions, such as in the air or underwater.

C. Initiatives for Waterproofing of EMG Sensors

The following examples and Table I summarize the previous research related to the waterproofing of EMG sensors.

Iizuka [9] waterproofed dry-type Ag/AgCl electrodes by covering them with a waterproof film. COMETA Systems [10] waterproofed disposable electrodes by applying a waterproof film and using a completely waterproof circuit.

These studies enabled the measurement of EMG signals underwater by waterproofing existing electrodes through the attachment of a waterproof film to the skin. However, this film can irritate the skin. Moreover, with this approach, the user must replace the electrodes and film each time they use the prosthetic hand, which can increase costs.

Duan et al. [11] performed waterproofing by covering a conductive silicone electrode with insulators, excluding the

EMG measurement surface, to enable myoelectric prosthetic hand users to perform underwater activities. They confirmed that the amplitude and frequency characteristics in air and water obtained by their sensors were equivalent to those obtained in air using conventional metal electrodes.

Ohtsu et al. [12] solved the problem of skin irritation due to an adhesive waterproof film, which was used during underwater EMG signal measurements, by using waterproofed insulated electrodes and an amplifier with high-input impedance instead of the film. They covered the electrodes of a conductive cloth with vinyl chloride and measured the EMG signal based on the capacitance between the electrodes and skin. The proposed sensor was used to investigate the attenuation rate of integrated EMG signals owing to the difference between air and water.

The proposed sensor acquired EMG signals in air similar to those acquired by a conventional metal electrode. The attenuation rate of the developed sensor was lower than that of the conventional metal electrode, which had an attenuation rate of 80% in water. Therefore, their proposed sensor could acquire similar signals in air and water.

These studies employed a flexible material for the cover to insulate and protect the electrodes. The electrodes deform and conform to the shape of the body, thereby improving the adhesion between the skin and electrodes, which has enabled the measurement of EMG signals in water. However, these studies only performed basic verifications in which the electrode covered by insulators creates the same environment as in air by adhering to the skin. Therefore, the relationship between the sensor and peripheral system was not considered. Furthermore, to make the sensor waterproof, a measurement system must be constructed by considering the pressing method to maintain adhesion between the skin and the electrode.

Myoelectric sensors made of flexible materials developed by our research group are described below. Jiang et al. [7] investigated the signal quality of sensors using nonwoven fabrics coated with polypyrrole as electrodes. The fabricated electrode showed a higher signal amplitude and lower noise than the conventional metal electrode owing to its flexible material and increased contact area with the skin. However, a polypyrrole electrode cannot measure signals when it absorbs sweat and is vulnerable to external forces.

Therefore, Jiang et al. [8] proposed a hybrid electrode with metal wires and conductive silicone, in which both contact the skin and the conductive silicone contacts the skin, and investigated the robustness of the signals owing to perspiration and external forces applied to the sensor. The hybrid electrode obtained a signal similar to that of a metal electrode while maintaining a stable signal quality under conditions such as perspiration or external force because of the flexibility of conductive silicone.

Mouri et al. [4] addressed the problem of delamination caused by the difference in hardness between conductive silicone and metals by using flexible wires. Pick-and-place experiments were conducted using their proposed sensor. Compared to conventional metal electrodes, flexible electrodes enable the sensor to cope with the deformation caused by muscular contraction, thus enabling stable signal acquisition and improving the success rate of the task.

Togo et al. [5] investigated the optimization of carbon concentration for the stable measurement of EMG signals in sensors using conductive silicone. They showed that stable EMG signals correlated with the capacitance and proposed a two-layered electrode structure with different carbon concentrations to enable the stable acquisition of EMG signals.

These studies developed robust EMG sensors that employ conductive silicone electrodes with flexible and controlled carbon concentrations to improve skin-to-electrode adhesion and optimize capacitance against noise from environmental factors, such as muscular contractions, external forces, and perspiration. However, prosthetic hand users may experience problems when performing tasks involving contact with water, such as, bathing, or toileting. The prosthetic hand cannot be used in water, as water entering the circuit causes a short circuit. Similarly, water entering between the skin and the electrodes changes the gain, causing the myoelectric prosthetic hand to stop working properly. For this reason, it is necessary not only to waterproof the EMG sensors, but also to apply pressure to the electrodes to prevent water from entering between the skin and the electrodes.

In the proposed sensor, the area around the electrode was waterproof, and the electrode was pressed against the skin to maintain an in-air environment between the skin and electrode to measure EMG signals. In the flexible electrode, the material was changed from metal to conductive silicone, the contact state between the skin and electrode was stabilized, and the capacitance was optimized by the carbon concentration of the electrode to stabilize the EMG signal such that it could be obtained regardless of external disturbances. In this study, we combined waterproofing and optimization methods for EMG signals, fabricated a conductive rubber-type waterproof sensor, and proposed a pressing method and a corresponding peripheral mechanism to maintain adhesion to the skin.

In previous studies, the waterproofed sensor was fixed with a rubber band; thus, contact between the skin and sensor was limited to the electrodes, and pressure was concentrated locally. As a result, the electrodes were easily detached from the skin when it deformed owing to muscle bulging, resulting in water intrusion and deterioration of signal quality. Therefore, the conventional method did not provide sufficient pressure dispersion, and measurement stability was diminished. In contrast, this study developed an integrated sensor band that uses a mesh-based silicone sheet to achieve uniform pressure dispersion while ensuring that the entire sensor adheres to the skin, thereby preventing water intrusion and enabling stable EMG signal acquisition. This method allows the sensor band to adapt to changes in skin curvature and muscle expansion, and provides improved waterproofing and measurement stability compared to those of conventional methods.

D. Approach of the Integrated EMG Sensor Band

We propose an integrated EMG sensor band with a mesh-based silicone sheet. As noted above, waterproofing the interface between the skin and electrodes is important

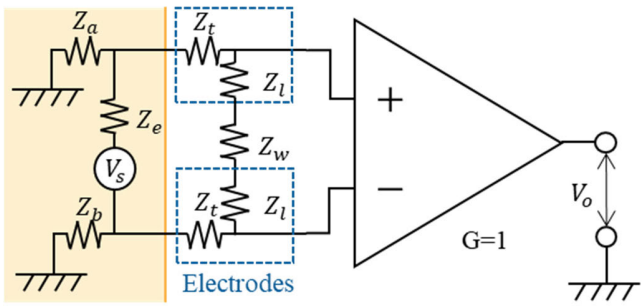

Fig. 1. Circuit model of the skin-electrode.

because impedance changes due to water ingress, resulting in a deterioration of the signal to noise ratio (SNR) of EMG signals. However, conventional studies on the waterproofing of EMG sensors do not consider maintaining an optimal environment around the sensor. The development of an integrated sensor band made of a silicone sheet prevents water from penetrating the interface between the skin and the electrodes and allows for even sensor loads. Initially, the bands had to be thicker because silicone is highly susceptible to tearing forces; however, by using a mesh as the base material, thin, light, and flexible bands could be produced. The flexibility of the bands also ensures that the load is always directed toward the center of the forearm, making stable measurement possible. The details of the production method and the characteristics of the developed sensor bands are described in Section II. The as-developed mesh-based silicone sheet has already been applied in EMG sensors [13].

II. Materials and Methods

A. Problem

The impedance characteristics of the electrode-skin interface exhibited notable differences between air and water. Referencing Rainoldi et al. [14], a circuit model of the skin-electrode interface was constructed (Fig. 1). Z_e and V_s represent the impedance and voltage derived from the muscle, respectively; Z_t and Z_l represent the transverse and longitudinal impedances of the electrode, respectively; and Z_w represents the electrode-to-electrode impedance. When measuring EMG signals in air, Z_w can be approximated as infinity. From the mechanism of bipolar EMG sensors, the condition $Z_a \approx Z_b \gg Z_e$ holds; therefore, if the gain of an amplifier is 1, the voltage ratio A air $(V_o - V_s)$ is 1. However, when measured in water, Z_w was smaller and similar to the others; therefore, the impact of Z_w could no longer be ignored. Considering an equivalent circuit, the voltage ratio $Awater$ can be expressed as follows:

$$Awater = \frac{V_o}{V_s} = \frac{Z_w + 2Z_l}{Z_w + 2Z_l + 2Z_t + Z_e} < 1 = Aair. \quad (1)$$

Note that $Awater$ will always be less than 1, and if Z_w is infinite, the voltage ratio will be 1. This indicates that measuring the EMG signals in water results in signals weaker than those measured in air.

B. Structure of the Proposed Mechanism

In this study, we proposed a band-type mechanism in which a surface EMG (sEMG) sensor and sheet are integrated

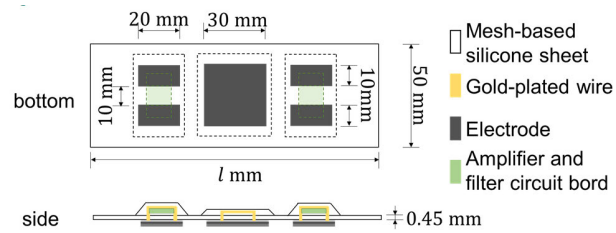

Fig. 2. Structure of the proposed mechanism.

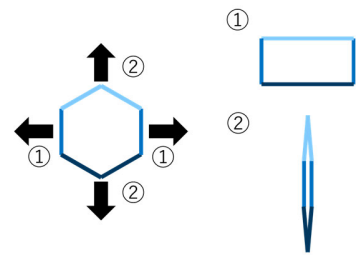

Fig. 3. Mesh sheet elongation model.

to prevent water ingress between the skin and electrode. Fig. 2 shows the structure of the proposed mechanism. This mechanism was fabricated by attaching electrodes to a mesh-based silicone sheet made of the same material as the conductive silicone electrodes. The fabrication method is detailed in Section II-D. Waterproofing between the skin and electrodes was provided by the mesh-based silicone sheet adhered to the skin. The entire band behaves like the waterproof seal of a dry suit to keep it waterproof. In this construction, the load toward the center of the forearm was made to be uniform with respect to the contact surface because the electrode was embedded in the band.

Theoretically, the load is uniquely determined by the diameter of the forearm, natural length of the sensor, and tensile length of the sensor. If the load that allows waterproofing between the skin and electrode is experimentally known, we can determine the conditions for the suitability of the proposed sensor for an unknown forearm diameter on the table.

This measurement system is capable of using many sensors simultaneously. Therefore, the placement of EMG sensors is only limited by the size of the silicone sheet and forearm circumference. We have produced a 3CH version and confirmed its operation. In this study, we describe the production of a 2CH version as an example to verify the quality of the EMG signal.

C. Selection of Silicone Sheet

Silicone is vulnerable to tearing forces. To solve this problem, we aim to improve the tearing strength of silicone sheets while maintaining their properties by filling a skeletal frame with silicone. In this study, three types of stretch fabric were selected as skeletal frames: elastic, spandex, and mesh sheets. Elastic and spandex bandages have sufficient elasticity and are used for medical purposes. The mesh sheet was a fabric with regularly arranged hexagonal holes, and the amount of stretching differed in the tensile direction, as shown in Fig. 3. Table II shows the characteristics of the frames filled

TABLE II
CHARACTERISTICS OF FRAMES

Material	Elastic bandage	Spandex bandage	Mesh sheet
Texture			
Cross section			
Thickness (cloth) [mm]	0.5	0.8	0.1
Thickness (cloth and silicone) [mm]	1.95	1.65	0.4
Weight [g / 100 mm^2]	0.182	0.231	0.036
Elongation rate [%]	240	270	260

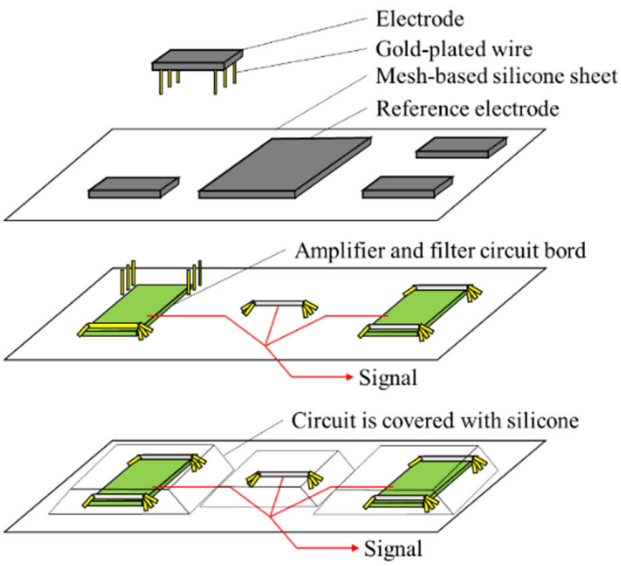

Fig. 4. Fabrication method.

with silicone. The elastic bandage had a good elongation rate before being filled with silicone; however, when filled with silicone, the thickness and weight of the bandage increased owing to surface irregularities, resulting in a decrease in the elongation rate. The spandex bandage was too fine to sufficiently fill the silicone, and the silicone and bandage were separated during stretching. Although the mesh sheet has a lower elongation rate, the material itself is very thin and has no unevenness. As a result, silicone can be filled into the sheet very thinly. Therefore, we decided to use a mesh sheet as a stretchable fabric to fill the silicone. Silicone sheets fabricated by filling a mesh sheet with silicone are known as mesh-based silicone sheets. Mesh-based silicone sheets can be processed with a minimum thickness of 0.45×10^{-3} m.

D. Fabrication Method

As shown in Fig. 4, the sensor band of the proposed mechanism was produced as follows.

1) Six wires attached to the electrodes were inserted into a mesh-based silicone sheet. This process was performed using two electrodes per amplifier board and one electrode as the reference electrode. The sheets and electrodes were secured using silicone between them.

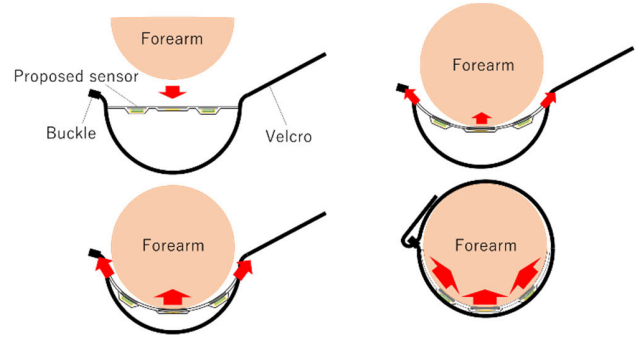

Fig. 5. Wearing method.

2) The six wires penetrating the sheet were bundled into bunches of three wires per side and soldered onto an amplifier board. The reference electrode was soldered to: a) the amplifier board by bundling the two wires into a single bunch. b) Subsequently, the area around the wires of the amplifier board and the reference electrode was covered with silicone to prevent water from penetrating.

E. Noise Filtering

Noise filtering of the EMG signals acquired by the sensors was performed using two processes: circuitry and software. In the circuitry, noise was removed using a 1–1000 Hz bandpass filter and a 50 Hz notch filter. The 50 Hz noise filter was used because this is the frequency of a commercial power supply in eastern Japan. The software also performed a frequency analysis using a 3.125 Hz high-pass filter, a 40–60 Hz band-stop filter, and a 400 Hz low-pass filter to reduce noise. These frequencies are based on the characteristics of the EMG signals.

F. Wearing Method

If the contact impedance between the skin and electrode is not constant, the amplification can easily change. Therefore, an equal load must be applied to all the electrodes. In this study, this problem was resolved by wrapping a band around the forearm. As shown in Fig. 5, by stretching the silicone sheet with the electrodes and PCB boards like a bowstring around the band, a force was uniformly generated in the direction of the center of the forearm according to the tension of the sheet, thus achieving the same contact impedance and constant amplification.

G. Calculation of Force on the Forearm

The constant loads associated with the forearm diameter, natural sheet length, and arc length were derived as follows: The forearm is assumed to be an ideal cylindrical object without gravity. The forearm radius is R[m], the sheet natural length is l[m], the arc length is L[m], the rubber elasticity of silicone is E[N/m^2], the cross-sectional area perpendicular to the elongation direction of silicone is S_0[m^2], and the derived mechanical model is shown in Fig. 6. When the sheet is elongated from its natural length l to the arc length L, the strain ε of the sheet can be expressed by

$$\varepsilon = \frac{(L - l)}{l}. \tag{2}$$

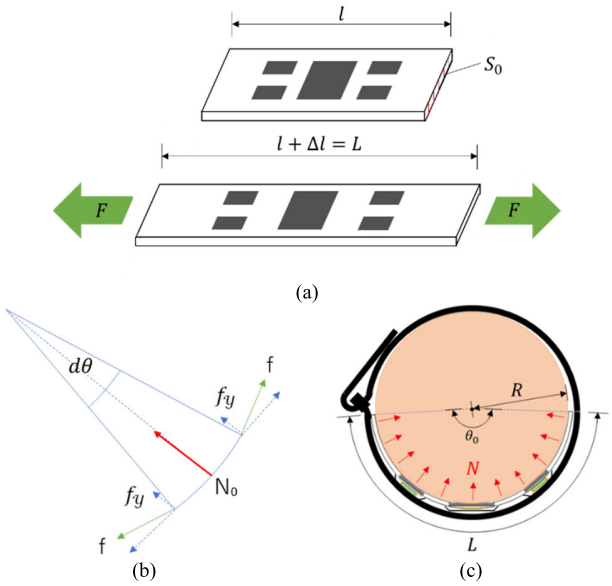

Fig. 6. Dynamics model. (a) elongation change, (b) small angle, and (c) proposed mechanism equipped on the forearm.

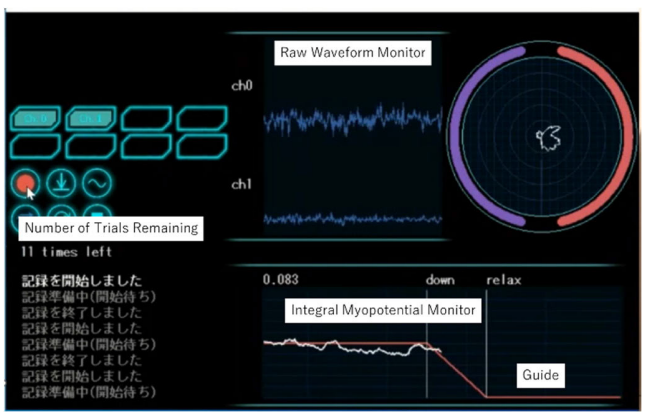

Fig. 7. Visual feedback system.

Therefore, the tensile strength of the sheet can be expressed as follows:

$$F = \varepsilon E S_0 = \frac{(L-l)}{l} E S_0. \tag{3}$$

Because the tensile strength F is similar for small lengths, the composite of the forces normal to the tensile strength F is equal to the centripetal force, given that the contact occurs at a small angle $d\theta$, as shown in Fig. 6(b), where the normal component F_y of F is

$$F_y = F \sin \frac{d\theta}{2} \tag{4}$$

and the centripetal force N_0 can be defined as below because when $\sin d\theta \to 0$, $\sin d\theta/2 d\theta/2$

$$N_0 = 2F_y = F d\theta = \frac{(L-l)}{l} E S_0 d\theta. \tag{5}$$

Considering θ_0 satisfying $L = R\theta_0$ as shown in Fig. 6(c), the load N acting in the range of the sheet contact toward the arm center is

$$N = \int_{d\theta}^{\theta_0 + d\theta} \frac{(L-l)}{l} E S_0 d\theta = \frac{L(L-l)}{Rl} E S_0. \tag{6}$$

This value is theoretical; however, the actual value can be obtained using a force sensor to determine the corresponding relationship.

III. Experiments

A. Load Variation With Extension

The proposed mechanism achieves waterproofing by pressing a mesh-based silicone sheet against the skin to prevent flooding between the electrode and the skin. The proposed mechanism is a band system, and the elongation of the mesh-based silicone sheet increases the pressure between the skin and mesh-based silicone sheet, which may improve the waterproofing of the sensor. We aim to use the theoretical (6) as a guide for determining the natural and arc lengths of the sensor for an unknown forearm diameter by experimentally showing that the theoretical loads are close to the measured ones for different arc lengths and forearm diameters.

1) Measurement System: A pressure-sensitive sensor (Interlink Electronics, Inc., FSR400) was used to measure the loads. A microcontroller (Arduino, Arduino Uno) was used to process the information. The data acquisition was performed at a sampling rate of 10 Hz and a quantization bit rate of 10 bits. A 10 kΩ resistor was connected to the pressure sensor.

2) Experimental Procedure: A pressure-sensitive sensor was placed in a cylinder (circumferences: 19.0, 24.0, and 29.4 cm), and a sheet deformed at six different lengths (0–5 cm) was wrapped around the cylinder, such that the reference electrode of the proposed mechanism contacted the pressure-sensitive sensor. The data for 10 s were acquired five times for each deformation.

3) Data Processing: The acquired data were converted to 0–5 V and the pressure conversion was performed according to the FSR400 datasheet. The acquired data were calculated using (7). The constant coefficient C is determined by the size of the resistors constituting the circuit and individual differences among the sensors. In this experiment, $C = 1$ was used, which was assumed as equivalent to the actual measured value of the load

$$N = \exp\left(\frac{V_{\text{out}}}{C}\right). \tag{7}$$

B. Flooding Experiment

The proposed mechanism achieves waterproofing of the interface between the skin and electrode by pressing a mesh-based silicone sheet against the skin with a load. When the pressing load was small, the silicone was not watertight, and the interface between the skin and the electrode was flooded. It was assumed that the impedance between the electrodes collapsed and that the measured signals in air and water were different. Under the above assumptions, sEMG measurements using the proposed mechanism were performed in air and water, and the measured signals were compared. In this experiment, we evaluated the sEMG signals using their SNR, median frequency (MDF), and attenuation rate.

1) Participant Information: The participants were four non-disabled males in their 20 s (Participants A, B, C, and D). This study was approved by the University of Electro-Communications Ethics Committee for Human Experiments [No. 10006(7)]. All the participants provided informed consent.

2) Measurement System: EMG signals were obtained from the EMG sensor input to an A/D converter (CONTEC Company Ltd., AIO-160802AY-USB). Data acquisition was performed at a sampling rate of 1.6 kHz and quantization bit rate of 16 bits. The A/D converter measurement range was −10 to 10 V. The output voltage of the EMG sensor used in this study ranged from 0 to 5 V, so the measurements performed using the A/D converter did not span the full range. To reduce interference from commercial power supplies, power was supplied using a mobile battery.

3) Experimental Procedure: The reference electrode of the proposed mechanism was placed in contact with the dorsal ulna 5 cm from the elbow, and the proposed mechanism was wrapped around the forearm. However, due to the nature of the experiment, the positions of the electrodes other than the reference electrode shifted when the sensor band was pulled. The EMG signals tended to change and were affected by the noise caused by moving hand postures. Therefore, the task order was set such that the transition periods between resting and grasping were excluded from the measurement. The task order was rest, transition period, grasp, and transition period, each lasting 5 s and repeated 40 times. This task was performed in both air and water. The band stretch changed in the order of 5, 3, and 1 cm, respectively. The longer the stretch change, the higher the pressure applied to the sensor. Visual feedback was used to control the magnitude of the EMG signals by displaying the mean absolute value (MAV) on the screen in real time (Fig. 7).

a) Signal to noise ratio: We used the SNR to evaluate the signal quality of the proposed sensors.

i) Evaluation function: The SNR is an indicator that measures the proportion of noise contained in the acquired signal, and is expressed by (8). $\hat{E}_{\text{grasp}}$ is the acquired signal, which is the root mean square (rms) of the signal during grasping, and $\hat{E}_{\text{relax}}$ is the noise signal, which is the rms of the signal during rest

$$\text{SNR} = 10 \log \frac{\hat{E}_{\text{grasp}}}{\hat{E}_{\text{relax}}}. \tag{8}$$

b) Median frequency: The MDF value differs depending on the muscle fiber type. We adopted the MDF as the muscle activity evaluation index because the same frequency can be measured if the same location is measured in both air and water environments.

i) Signal processing: The MDF is a frequency-based evaluation method; therefore, EMG signals must be analyzed using frequency analysis. A fast Fourier transform (FFT) was performed on the EMG signals to transform the frequency. The integral interval was 512 data points, and the FFT was performed in a sliding manner for each data point. A Hamming window was used as the window function.

TABLE III
INTERNATIONAL PROTECTION

IP code	Summary
0	Not rated for protection
1	Protection against vertically falling water droplets (condensation)
2	Protection against vertically falling water droplets when tilted at 15°
3	Protection against moisture spray at up to 60°
4	Protection against splashing water from any direction for a minimum of water
5	Protection against low-pressure jets of water
6	Protection against powerful jets of water at a rate of 100 L/min
7	Protected against full immersion up to 15–100 cm in depth for 30 min
8	Protection against full immersion up to 1 m or more in depth for more than 30 min

ii) Evaluation function: The MDF was used to determine the frequency; this value is the median when the absolute squared power spectral density [psd(f)] is considered as a histogram. The MDF (f_{md}) is expressed by (9). f is the frequency after applying the FFT to the measured EMG signals. In this experiment, the measurement range of the EMG frequency was 3.125–400 Hz; the integration ranges were $3.125 - f_{\text{md}}$ and $f_{\text{md}} - 400$. The value of f_{md} was determined when the value of psd(f) was equal for both values

$$\int_{f_{\text{md}}}^{400} \text{psd}(f)df = \int_{3.125}^{f_{\text{md}}} \text{psd}(f)df. \tag{9}$$

c) Attenuation rate: The amplification factor of a myoelectric sensor varies depending on the magnitude of the impedance between the electrodes and the contact resistance. Therefore, water ingress between the skin and electrodes changes the signal amplification rate, and the signal acquisition is difficult to stabilize owing to noise. The attenuation rate of the signal in water compared to that in air was measured to evaluate the effect of water ingress.

i) Evaluation function: $\hat{E}$ is the measured rms of the voltage E. The attenuated ratio ζ_i is the ratio of the rms of the ith measured underwater rms, $\hat{E}_{\text{water}_i}$, to the average rms of 40 trials in air, $\hat{E}_{\text{land}}$

$$\hat{E} = \sqrt{\frac{1}{M}\sum_{n=0}^{M}|E_n|^2} \tag{10}$$

$$\zeta_i = \frac{\hat{E}_{\text{land}} - \hat{E}_{\text{water}_i}}{\hat{E}_{\text{land}}} \times 100. \tag{11}$$

C. IP Test

The use of a prosthetic hand can expand the range of ADLs for amputees. ADLs include activities that involve contact with water, such as bathing and cooking. The Japanese Industrial Standards (JISs) established the international protection (IP) test and waterproofing rating for electronic components (Table III). This standard was used to verify the waterproofing performance of the proposed mechanism. First, we tested an IPX4 rating to verify the waterproofing performance of the shower during bathing. In addition, we conducted IPX 7, the most stringent IP test, to verify the strength of waterproofing.

1) IPX4 Flowing Water Test: The IPX4 test consists of "Water falling (10 L/min) within ±180° from the vertical. The test duration is 5 min." We conducted an experiment to satisfy this condition using the following process.

1) The A/D converter was placed outside the bathroom to avoid getting wet.
2) The flow rate was adjusted to be at least 10 L/min.
3) The proposed mechanism with an extension of 5 cm was attached to the participant's forearm.
4) The camera and sEMG measurements were synchronized and recording was started.
5) The operator sprayed water on the forearm at various angles, and the participant performed three movements: resting, grasping, and opening.
6) The measurements were continued for at least 5 min and repeated five times.

2) IPX7 Immersion Test: The IPX7 test consists of "immersion in water 15–100 cm for 30 min." We conducted an experiment to satisfy this condition using the following process.

1) The container was filled with water to a level of 40 cm from the bottom.
2) The proposed mechanism was submerged in the bottom of the container to prevent the A/D converter connection from getting wet.
3) After 30 min, the sensor was pulled up and wiped lightly around the sensor.
4) We confirmed that the proposed mechanism was capable of measuring muscle activity.

IV. Results

A. Load Variation With Extension

The measured pressure values increased monotonically with the stretching length of the proposed mechanism regardless of the circumference of the cylindrical object to which the proposed mechanism was attached. Therefore, if the load between the proposed mechanism and the skin is increased, the stretch from the natural length should be increased.

We verified whether the measured value of the load corresponds to the theoretical value defined in (6). We can obtain the model values for the stretching length $L - l$ and load N by using the theoretical equation and the natural length of the band $l = 14.0$ cm for the proposed mechanism in this experiment. The pressure trends were compared by comparing the change in pressure with the amount of stretching, although the vertical axis scales of the measured and theoretical values were different.

Fig. 8(a) shows the theoretical and measured pressures on cylindrical objects with circumferences of 19.0, 24.0, and 29.4 cm. The measured values in Fig. 8(b) are the average and standard deviation of five measurements.

The theoretical values showed a monotonic increase in proportion to the circumference, whereas the measured values changed in order at an elongation of 1 cm. The trend after a 2 cm stretch is the same as the theoretical value: the smaller the circumference, the greater the increase in load with the amount of stretching. This is because the output voltage obtained by the pressure-sensitive sensor increases logarithmically with the load and the amount of stretching is small; therefore, the error is large when the load is small.

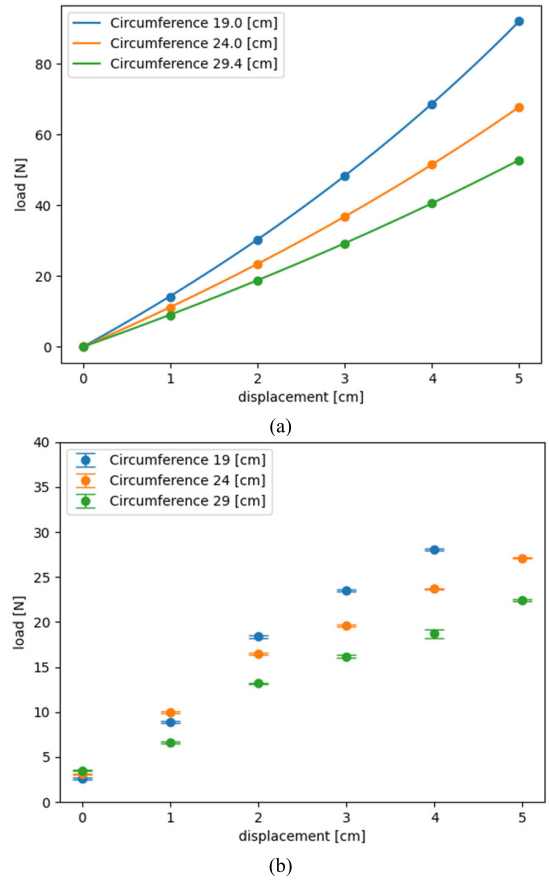

Fig. 8. Relationship between sensor elongation change and force applied to the forearm. (a) Theoretical value and (b) measured values.

When the amount of stretching was 0 cm, the measured values did not include the load from band tension. These values indicate the mass of the proposed mechanism, because the pressure-sensitive sensor was placed vertically upward.

The trend of the measured values after stretching for 2 cm was close to the trend of the theoretical value, and the values could be approximated using a linear or quadratic equation. The theoretical equation is quadratic, and the force acting on a cylindrical object increases infinitely as the stretching length increases. Due to the structure of the proposed mechanism, the maximum value of the sensor band length L is the forearm circumference $2\pi R$, so it is not necessary to consider using a length longer than that tested in this experiment.

In this experiment, we compared whether the values corresponding to the measured values of the loads correspond to the theoretical equations. Although quantitative comparisons are difficult because of the different scales, we found that the values corresponding to the measured values of the load were close to the theoretical values.

This means that when a value corresponding to the measured value of the load that satisfies the waterproofing between the skin and electrode is obtained, this value also corresponds to the model value. In turn, this suggests that the conditions that satisfy the waterproofing between the skin and electrode can be derived from the natural length of the

TABLE IV
SIGNAL TO NOISE RATIO

Amount of stretching: 1 cm

participant	relax		grasp		SNR [dB]	
	Air	Water	Air	Water	Air	Water
A	$8.09\times10^{-3}(6.95\times10^{-4})$	$7.17\times10^{-3}(1.15\times10^{-3})$	$9.38\times10^{-2}(4.95\times10^{-3})$	$9.31\times10^{-2}(7.21\times10^{-3})$	10.6	11.1
B	$1.19\times10^{-2}(1.06\times10^{-2})$	$1.58\times10^{-2}(4.89\times10^{-3})$	$1.87\times10^{-1}(1.18\times10^{-2})$	$2.05\times10^{-1}(1.53\times10^{-2})$	12.0	11.1
C	$2.17\times10^{-2}(9.55\times10^{-3})$	$7.37\times10^{-2}(5.09\times10^{-2})$	$2.18\times10^{-1}(3.01\times10^{-1})$	$1.78\times10^{-1}(2.41\times10^{-2})$	10.0	3.82
D	$9.64\times10^{-3}(1.60\times10^{-3})$	$8.64\times10^{-2}(4.65\times10^{-2})$	$1.08\times10^{-1}(8.27\times10^{-3})$	$3.30\times10^{-1}(4.08\times10^{-1})$	10.5	5.80

Amount of stretching: 3 cm

participant	relax		grasp		SNR [dB]	
	Air	Water	Air	Water	Air	Water
A	$8.20\times10^{-3}(6.86\times10^{-4})$	$8.57\times10^{-3}(1.86\times10^{-3})$	$9.80\times10^{-2}(5.85\times10^{-3})$	$8.53\times10^{-2}(1.74\times10^{-3})$	10.8	9.98
B	$8.63\times10^{-3}(2.06\times10^{-3})$	$1.05\times10^{-2}(2.89\times10^{-3})$	$1.94\times10^{-1}(2.65\times10^{-2})$	$1.83\times10^{-1}(2.45\times10^{-2})$	13.5	12.4
C	$6.90\times10^{-3}(4.48\times10^{-4})$	$6.39\times10^{-3}(5.34\times10^{-4})$	$2.22\times10^{-1}(2.48\times10^{-2})$	$2.02\times10^{-1}(2.16\times10^{-2})$	15.1	15.0
D	$1.17\times10^{-2}(6.15\times10^{-3})$	$7.16\times10^{-3}(1.01\times10^{-3})$	$9.94\times10^{-2}(8.27\times10^{-3})$	$1.06\times10^{-1}(1.09\times10^{-2})$	9.30	11.7

Amount of stretching: 5 cm

participant	relax		grasp		SNR [dB]	
	Air	Water	Air	Water	Air	Water
A	$8.89\times10^{-3}(9.85\times10^{-4})$	$7.12\times10^{-3}(1.89\times10^{-3})$	$9.34\times10^{-2}(8.42\times10^{-3})$	$1.01\times10^{-1}(9.35\times10^{-3})$	10.2	11.5
B	$1.37\times10^{-2}(7.21\times10^{-3})$	$1.04\times10^{-2}(5.13\times10^{-3})$	$1.27\times10^{-1}(1.34\times10^{-2})$	$1.60\times10^{-1}(2.42\times10^{-2})$	9.70	11.9
C	$6.92\times10^{-3}(2.18\times10^{-4})$	$7.49\times10^{-3}(1.28\times10^{-3})$	$1.14\times10^{-1}(1.57\times10^{-2})$	$1.02\times10^{-1}(1.62\times10^{-2})$	12.2	11.3
D	$7.38\times10^{-3}(5.13\times10^{-4})$	$7.54\times10^{-3}(8.56\times10^{-4})$	$8.60\times10^{-2}(7.12\times10^{-3})$	$1.04\times10^{-1}(1.31\times10^{-2})$	10.7	11.4

TABLE V
MEDIAN FREQUENCY

Amount of stretching: 1 cm

participant	relax [Hz]			grasp [Hz]		
	Air	Water	1-p	Air	Water	1-p
A	193.1(3.36)	194.9(4.45)	6.18×10^{-3}	189.7(3.22)	184.0(3.38)	6.32×10^{-10}
B	187.1(14.2)	179.2(10.6)	7.00×10^{-4}	212.1(2.84)	206.3(4.48)	9.57×10^{-9}
C	202.0(5.09)	202.8(30.2)	4.34×10^{-2}	195.4(19.8)	203.5(7.00)	7.45×10^{-2}
D	190.9(6.22)	203.4(24.6)	4.56×10^{-3}	203.0(2.45)	180.5(24.5)	2.71×10^{-8}

Amount of stretching: 3 cm

participant	relax [Hz]			grasp [Hz]		
	Air	Water	1-p	Air	Water	1-p
A	195.1(3.44)	180.8(14.5)	1.19×10^{-7}	196.8(2.24)	188.2(6.34)	3.25×10^{-10}
B	190.5(7.47)	178.3(13.5)	8.41×10^{-5}	212.3(4.95)	209.5(5.19)	2.02×10^{-2}
C	194.2(4.21)	199.0(3.52)	4.61×10^{-7}	189.2(5.25)	191.5(3.39)	2.65×10^{-2}
D	200.5(3.62)	196.9(4.25)	8.74×10^{-5}	193.5(2.76)	189.0(4.59)	1.27×10^{-6}

Amount of stretching: 5 cm

participant	relax [Hz]			grasp [Hz]		
	Air	Water	1-p	Air	Water	1-p
A	186.2(3.79)	186.9(7.69)	1.66×10^{-1}	205.7(2.85)	200.0(4.71)	5.28×10^{-10}
B	193.5(9.50)	184.4(14.1)	2.01×10^{-3}	213.3(2.86)	207.5(3.57)	2.31×10^{-9}
C	194.1(4.25)	192.2(8.41)	3.02×10^{-1}	187.9(5.29)	190.6(5.43)	5.62×10^{-2}
D	198.8(4.36)	196.0(6.92)	1.52×10^{-2}	191.1(2.92)	206.5(3.68)	6.88×10^{-15}

as-developed sensor, extension length, and amputation limb radius. Therefore, a theoretical value can be obtained when creating a development mechanism adapted to an unknown forearm defect radius.

B. Flooding Experiment

1) Signal to Noise Ratio: Table IV lists the rms and SNR values for each stretch length. Values in parentheses indicate standard deviations. Water ingress was observed when the elongation amount was 1 cm for participants C and D. The SNR values for the group where no water ingress was observed were confirmed to be 9.3 dB or more. The cases in which water ingress was observed were confirmed to have SNR values of 5.8 dB or less. When water ingress occurred, the ratio of noise increased by 1.5 times. This increase is thought to be due to the fact that, as shown by (1), the impedance between the electrodes decreases and the gain increases when the sensor is immersed in water. In addition, when grasping, the sensor band's internal pressure increased due to the muscle bulging, and the water was pushed out, allowing the SNR in the water to obtain the same gain as in the air. However, since no muscle bulging occurred at rest, the gain fluctuated and the SNR deteriorated. The data without water ingress showed that the signal was being effectively acquired against the noise, as the ratio of noise was approximately 10%.

2) Median Frequency: Table V shows the median frequencies for each stretch length. Values in parentheses indicate standard deviations. We tested the identity of the MDF under each condition using a nonparametric Wilcoxon rank-sum test. The 1-p indicates the p-value subtracted from 1, which is the result of the test. The test results indicated no significant differences in any of the data. Therefore, the median frequencies in air and water showed no differences, confirming that the data were identical. These results indicated that the quality of the signals in air and water was the same.

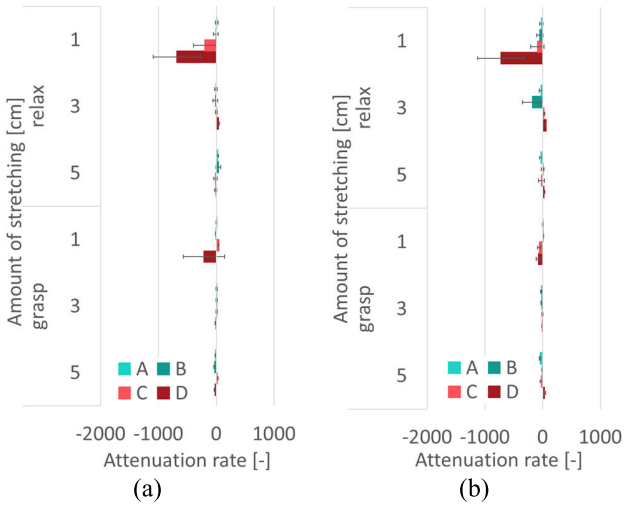

Fig. 9. Relationship between sensor elongation change and attenuation rate. (a) Sensor 1 and (b) sensor 2.

TABLE VI
NUMBER OF TIMES $|\zeta| > 50\%$ APPEARS

	A	B	C	D
$\lvert\zeta\rvert > 50$	17	76	11	49

3) Attenuation Rate: Four subjects participated in the experiment: two (Participants C and D) had water ingress between the skin and the electrode at 1 cm of stretching. Fig. 9 shows the average attenuation rate for each rms value classified according to the presence or absence of water ingress between the skin and the electrode.

The attenuation rate without immersion was within $\pm 50\%$ for all amounts of stretching except for the 3 cm relaxation of sensor 2 for Participant B. Therefore, the sensor had a similar gain in the absence of water immersion. In (1), the resistance between electrodes is defined as Z_w. If there is no water intrusion, Z_w can be considered infinite, and the gain is 1. Therefore, the same amplification is considered to be the output if there is no water intrusion and the attenuation rate is considered small. Thus, when the amount of stretching was greater than 3 cm, water inundation was considered absent.

The attenuation rate of -180% at 3 cm may be due to the following reasons. Participant B may have been unable to maintain a resting state while acquiring the EMG. The number of occurrences of an attenuation rate of $\pm 50\%$ or less was almost twice that of the other participants, except for the 1 cm of water immersion (Table VI). As this condition occurred frequently at 3 cm of relaxation in water, the attenuation rate was low, indicating that the force could not be relaxed. In addition, the standard deviation of the resting state without water immersion was larger because Participant B had difficulty relaxing at rest.

The attenuation ratio of sensor 1, where water immersion occurred when the stretching was 1 cm, was -500% at rest and -70% during grasping, indicating that the attenuation ratio was increasing. According to (1) of the differential amplifier, when water enters between the skin and electrode, the gain increases owing to the decrease in the resistances of Z_w and Z_t. Therefore, the gain increased with water immersion compared with that in air, which resulted in a larger signal. When immersed in water, the contact impedance between the skin and the electrode was unstable and could not maintain the same gain, which resulted in a larger variance. The attenuation rate during the grip was larger than that at rest because the circumference of the forearm increased during grasping, and the water between the skin and electrode was removed by the pressure.

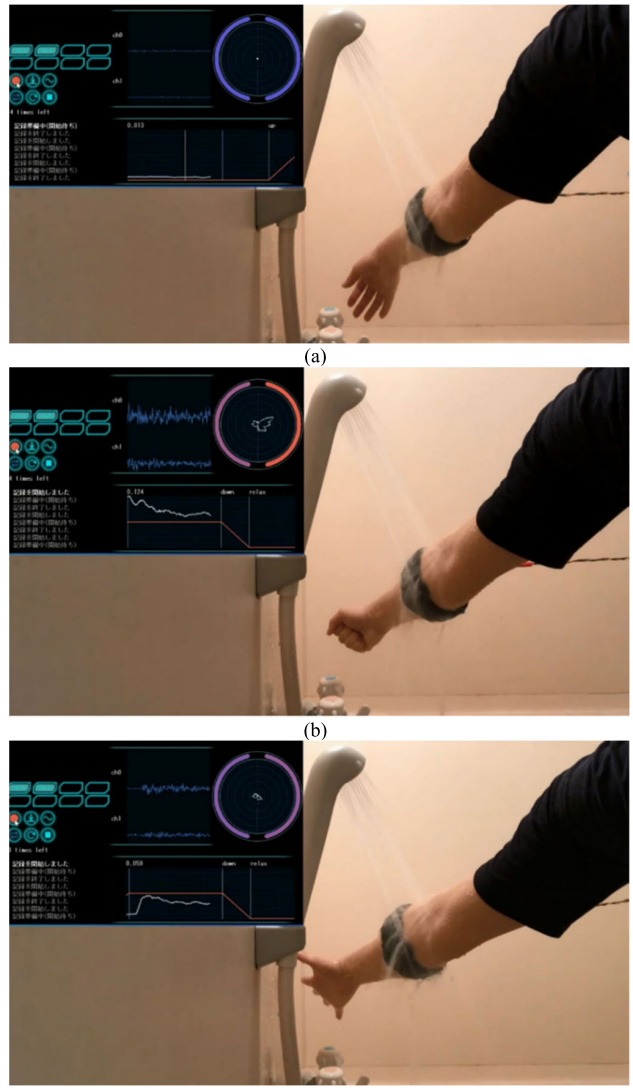

Fig. 10. Measurements during IPX4 testing. (a) Rlaxation, (b) grasping, and (c) opening.

Therefore, the change in gain due to the contact resistance is considered to have been reduced. However, as shown in Fig. 9(b), the presence of a step between the mesh-based silicone sheet and the electrodes did not reduce water ingress between the electrodes, regardless of the circumferential diameter change of the forearm. This reduced the noise derived from water ingress between the skin and the electrodes, but did not reduce the water ingress between the electrodes, which may have caused variance due to noise.

These results indicate that applying a certain amount of pressure to the proposed mechanism prevents ingress, and that the signal quality is the same under both conditions.

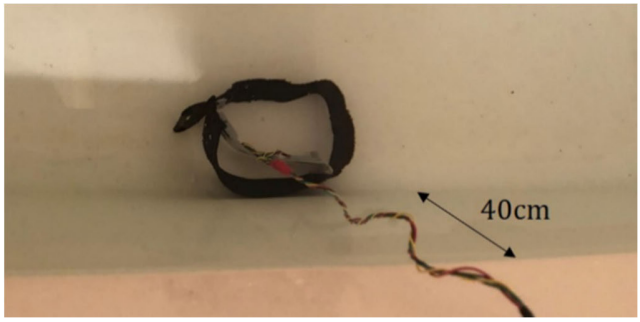

Fig. 11. Experimental environment of IPX7 test.

C. IP Test

1) IPX4 Flowing Water Test: Even when the proposed mechanism was subjected to the IPX4 test, the sensor was capable of measuring EMG signals when the resting, opening, and closing movements of the hand were performed (Fig. 10).

2) IPX7 Immersion Test: Even when the proposed mechanism was submerged in a bathtub filled with 40 cm of water (Fig. 11), the sensor could measure EMG signals when resting, opening, and closing hand movements were performed.

V. Discussion

A. Load Variation With Extension

The load change when the proposed mechanism was wrapped around the cylinder exhibited the same tendency as in the theoretical equation, and we confirmed that the load increased with the elongation of the proposed mechanism. Therefore, a large load can be applied to the sensor by stretching the band, thereby improving the degree of adhesion between the sensor and skin. However, in this study, the theoretical and measured scales differed and the theoretical equation did not consider the weight of the sensor.

B. Flooding Experiment

1) Signal to Noise Ratio: The proposed mechanism was confirmed to have a noise ratio of approximately 10% when a certain amount of pressure was applied to the sensor both underwater and in air. Therefore, we confirmed that there was no deterioration in signal quality when the sensor was under load.

2) Median Frequency: It was confirmed that the proposed mechanism did not change its MDF in either air or water. Therefore, low- and high-frequency noises were not generated, and the signal quality was maintained for the proposed mechanism in either environment.

3) Attenuation Rate: We confirmed that when a constant pressing pressure was applied to the sensor, the EMG signal in water had the same signal strength as that in air; thus, the signal attenuation was small. Therefore, we confirmed that the signal strength was maintained in both environments when the proposed mechanism was used.

4) IP Test: As a result of the flooding experiments conducted in accordance with JIS standards, we confirmed that the proposed mechanism is capable of acquiring EMG signals. Therefore, the myoelectric sensor itself is waterproof, and even if the electrodes are exposed to water, signal acquisition is not affected.

C. Limitations of the Study and Future Work

Customization of the electrode position according to individual users could solve the problem of drift due to user variability. This customization would allow the use of prosthetic hand systems with threshold and proportional control functions. Although the proposed sensor system is covered with silicone and waterproof, the gain cannot be adjusted according to the strength of the EMG signal. As the quality of the EMG signal changes, the electrode positions must be adjusted, which is difficult to do with the proposed system. However, control technology using pattern recognition allows software to process differences in individual muscle activity and enables the user to effectively operate the prosthetic hand. Because the signal input is independent of the magnitude and quality of the EMG signal, changes in the quality of the EMG signal have little effect on the operability of the prosthetic hand. This approach would allow the proposed sensor system to be used more effectively.

The EMG signals of individuals with acquired forearm defects differ from those of nondisabled individuals. Therefore, the frequency characteristics may change. In this study, the proposed sensor was applied only to nondisabled people; consequently, we were unable to validate the EMG signals for amputees. In addition, we were not able to verify the discrimination rate when the proposed sensor was used and the signal changed owing to the load applied to the sensor when a prosthetic hand was used. Moreover, if the proposed mechanism is wrapped around the forearm, it may not be possible to obtain sufficient pressure within the sensor band in people with forearm defects in which the forearm tapers toward the end. At present, we are planning to apply the proposed mechanism to silicone liners for use by people with forearm defects.

VI. Conclusion

Waterproof myoelectric prosthetic hands are important for enabling users to perform ADLs. Although waterproofing is also important for myoelectric sensors, attention has been focused only on the waterproofing of sensor circuits. In contrast, waterproofing of the contact surface between the sensor and the skin, where the signal is measured, has not been sufficiently considered.

This study aimed to develop a waterproof myoelectric sensor and propose a mounting method and peripheral mechanisms that consider the stabilization of the load balance of the electrodes. The proposed mechanism consisted of electrodes, a mesh-based silicone sheet, and an amplifier. The electrodes were then placed on the sheet. The degree of adhesion between the skin and electrodes could be changed by stretching the sheet to ensure waterproofing.

The performance of the proposed mechanism was evaluated in terms of the signal quality and waterproofing during ADLs.

We evaluated the signal in terms of the SNR, MDF, and attenuation rate in four participants under three different pressures in air and water environments. The proposed mechanism obtained similar EMG signals when evaluating the SNR, MDF, and attenuation rate. Therefore, the developed sensor can obtain the same EMG signals in air and water, indicating

that the sensor is waterproof. We conducted a waterproofing evaluation of the sensor in ADLs using a protection class test standardized by JIS. Exposure of the sensor to water for a certain period did not affect the myoelectric potential acquisition. Therefore, the proposed mechanism is waterproof, enabling users to perform their daily activities. Based on the above results, we developed a sensor that includes a peripheral mechanism that inhibits the penetration of water between the skin and electrodes, indicating that muscle activity can be measured even underwater.

In this study, the developed sensor was applied to nondisabled people to evaluate its waterproofing performance. However, because amputees use myoelectric prosthetic hands, we have not been able to conduct research under real-world conditions in which the prosthetic hand is equipped and the forearm muscle arrangement is different. Therefore, the as-developed sensor should be applied to amputees to evaluate the resulting waterproofing performance and signals.

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References

[1] A. Searle and L. Kirkup, "A direct comparison of wet, dry and insulating bioelectric recording electrodes," *Physiol. Meas.*, vol. 21, no. 2, pp. 271–283, May 2000, doi: 10.1088/0967-3334/21/2/307.

[2] N. Gandhi, C. Khe, D. Chung, Y. M. Chi, and G. Cauwenberghs, "Properties of dry and non-contact electrodes for wearable physiological sensors," in *Proc. Int. Conf. Body Sensor Netw.*, Dallas, TX, USA, May 2011, pp. 107–112, doi: 10.1109/BSN.2011.39.

[3] Y. Hiyoshi et al., "Development of a parent wireless assistive interface for myoelectric prosthetic hands for children," *Frontiers Neurorobot.*, vol. 12, p. 48, Aug. 2018, doi: 10.3389/fnbot.2018.00048.

[4] Y. Mouri et al., "Development of new flexible dry electrode for myoelectric sensor using conductive silicone," in *Proc. IEEE Int. Conf. Cyborg Bionic Syst. (CBS)*, Shenzhen, China, Oct. 2018, pp. 478–482, doi: 10.1109/CBS.2018.8612280.

[5] S. Togo, Y. Murai, Y. Jiang, and H. Yokoi, "Development of an sEMG sensor composed of two-layered conductive silicone with different carbon concentrations," *Sci. Rep.*, vol. 9, no. 1, p. 13996, Sep. 2019, doi: 10.1038/s41598-019-50112-4.

[6] J. M. Linacre, A. W. Heinemann, B. D. Wright, C. V. Granger, and B. B. Hamilton, "The structure and stability of the functional independence measure," *Arch. Phys. Med. Rehabil.*, vol. 75, no. 2, pp. 127–132, Feb. 1994.

[7] Y. Jiang, M. Togane, B. Lu, and H. Yokoi, "SEMG sensor using polypyrrole-coated nonwoven fabric sheet for practical control of prosthetic hand," *Frontiers Neurosci.*, vol. 11, p. 33, Feb. 2017, doi: 10.3389/fnins.2017.00033.

[8] Y. Jiang, Y. Murai, T. Kuwahara, S. Togo, Y. Yabuki, and H. Yokoi, "Conductive silicon based sEMG sensor for myoelectric control of prosthetic hands: Structure design," in *Proc. IEEE Int. Conf. Real-Time Comput. Robot. (RCAR)*, Okinawa, Japan, Jul. 2017, pp. 383–387, doi: 10.1109/RCAR.2017.8311891.

[9] S. Iizuka, "EMG analysis during front swimming," Ph.D. dissertation, Dept. Sport Sci., Waseda Univ., Tokyo, Japan, 2018.

[10] L. Puce et al., "Muscle fatigue and swimming efficiency in behind and lateral drafting," *Frontiers Physiol.*, vol. 13, Mar. 2022, Art. no. 835766, doi: 10.3389/fphys.2022.835766.

[11] F. Duan, J. Xue, Z. Sun, X. Chen, Y. Liu, and H. Yokoi, "A novel waterproof sEMG electrode based on conductive silicone for underwater signal recognition," *IEEE Sensors J.*, vol. 21, no. 18, pp. 20388–20397, Sep. 2021, doi: 10.1109/JSEN.2021.3095118.

[12] M. Ohtsu, Y. Fukuoka, and A. Ueno, "Underwater EMG measurement using a waterproof insulated electrode," *Adv. Biomed. Eng.*, vol. 1, pp. 81–88, Dec. 2012, doi: 10.14326/abe.1.81.

[13] I. Morimoto, Y. Yamanoi, M. Sugi, Y. Jiang, and H. Yokoi, "Development of integrated multi-electrode for FES using stretchable mesh material to induce hand posture," in *Proc. IFESS*, 2021, pp. 156–159.

[14] A. Rainoldi, C. Cescon, A. Bottin, R. Casale, and I. Caruso, "Surface EMG alterations induced by underwater recording," *J. Electromyography Kinesiol.*, vol. 14, no. 3, pp. 325–331, Jun. 2004, doi: 10.1016/j.jelekin.2003.10.002.

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